

Mathematics for Machine Learning

<https://mml-book.github.io/book/mml-book.pdf>

1 Linear Algebra

1.1 vectors and systems

- Polynomials form a vector space.
- Elements of $\mathbb{R}^n$ (tuples of n real numbers) are vectors. For example,

$$a = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \in \mathbb{R}^3.$$

- Adding two vectors $a, b \in \mathbb{R}^n$ component-wise results in another vector:
 $a + b = c \in \mathbb{R}^n$.
- Scaling: $a \in \mathbb{R}^n, \lambda \in \mathbb{R} \implies \lambda a \in \mathbb{R}^n$.
- Systems of equations: solution by contradiction,

$$x_1 + x_2 + x_3 = 3 \tag{1}$$

$$x_1 - x_2 + 2x_3 = 2 \tag{2}$$

$$2x_1 + 3x_3 = 1 \tag{3}$$

has *no solution*: Adding the first two equations yields $2x_1 + 3x_3 = 5$, which contradicts the third equation (3).

- Systems of equations: idk this is kinda basic but still,

$$x_1 + x_2 + x_3 = 3 \tag{1}$$

$$x_1 - x_2 + 2x_3 = 2 \tag{2}$$

$$x_2 + x_3 = 2 \tag{3}$$

$$(1) + (2) = 2x_1 + 3x_3 = 5, \text{ i.e., } x_3 = 1.$$

From (3): we then get that $x_2 = 1$. so $(1, 1, 1)$ is the only possible and *unique solution*

- Systems of equations: omission

$$x_1 + x_2 + x_3 = 3 \quad (1)$$

$$x_1 - x_2 + 2x_3 = 2 \quad (2)$$

$$2x_1 + 3x_3 = 5 \quad (3)$$

Since $(1) + (2) = (3)$, we omit the third equation. solve with (1) and (2) and let $x_3 = a \in \mathbb{R}$.

- Solving systems with vectors: let $a_{ij} \in \mathbb{R}$.

$$\begin{bmatrix} a_{11} \\ \vdots \\ a_{m1} \end{bmatrix} x_1 + \begin{bmatrix} a_{12} \\ \vdots \\ a_{m2} \end{bmatrix} x_2 + \cdots + \begin{bmatrix} a_{1n} \\ \vdots \\ a_{mn} \end{bmatrix} x_n = \begin{bmatrix} b_1 \\ \vdots \\ b_m \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & & \vdots \\ a_{m1} & \cdots & a_{mn} \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ \vdots \\ b_m \end{bmatrix}$$

1.2 matrices

1.2.1 Matrix Addition and Multiplication

Sum of Matrices $A, B \in \mathbb{R}^{m \times n}$ is defined as the element-wise sum, i.e.,

$$A + B := \begin{bmatrix} a_{11} + b_{11} & \cdots & a_{1n} + b_{1n} \\ \vdots & & \vdots \\ a_{m1} + b_{m1} & \cdots & a_{mn} + b_{mn} \end{bmatrix} \in \mathbb{R}^{m \times n}. \quad (2.12)$$

For matrices $A \in \mathbb{R}^{m \times n}$ and $B \in \mathbb{R}^{n \times k}$, the elements c_{ij} of the product $C = AB \in \mathbb{R}^{m \times k}$ are computed as

$$c_{ij} = \sum_{l=1}^n a_{il} b_{lj}, \quad i = 1, \dots, m, \quad j = 1, \dots, k. \quad (2.13)$$

This is the usual matrix multiplication.